Que Liu

https://que-liu.github.io/

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Education

University of Shanghai for Science and Technology **Sept 2021 – Jun 2026 (Expected)**

Bachelor's Degree in Intelligence Science and Technology *Shanghai, China*

- GPA: 3.52/4.00, **Major GPA**: 3.89/4.00
- Core Courses with scores: Machine Vision (92), Autonomous Mobile Robots (98), Machine Learning (93), Robot Vision System and Measurement (92), Natural Language Understanding (95), Intelligent Simulation (91)

Publications

1. Yunhao Xing, **Que Liu**, Jingwu Wang, and Diego Gómez-Zará. **"SMoRe: Spatial Mapping and Object Rendering Environment."** In *Companion Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25 Companion)*, 115–119. ACM, 2025. https://doi.org/10.1145/3708557.3716337

2. Muhammad Salman Abid, Mrigank Pawagi, Sugam Adhikari, Xuyan Cheng, Ryed Badr, Md Wahiduzzaman, Vedant Rathi, Ronghui Qi, Choiyin Li, Lu Liu, Rohit Sai Naidu, Licheng Lin, **Que Liu**, Asif Zubayer Palak, Mehzabin Haque, Xinyu Chen, Darko Marinov, and Saikat Dutta. **"GlueTest: Testing Code Translation via Language Interoperability."** In *Proceedings of the 40th International Conference on Software Maintenance and Evolution (ICSME'24) - NIER Track*, 2024.

Research Experience

Robot Position Data Transformation and Communication **Nov 2025 – Present**

*Graduation Thesis supervised by **Prof. Dongxiang Fu*** *University of Shanghai for Science and Technology*

- Designing and implementing position data transformation across static and dynamic coordinate frames in ROS (Robot Operating System), inspired by multi-task communication and kinematics applications.

Advanced Control Systems Lab **May 2025 – Present**

*Undergraduate Researcher supervised by **Prof. Andrea L'Afflitto*** *Virginia Tech*

- Applied genetic algorithms and multi-objective optimization to systematically tune gains in MRAC (Model Reference Adaptive Control), balancing tracking error, control effort, and stability.
- Designed and conducted sensitivity analysis of system parameters, identifying parameters effect in unmanned aerial vehicle's coupled dynamics.
- Integrated convex optimization based LMI (Linear Matrix Inequality) formulations to ensure the asymptotic stability of adaptive observers via the Kalman–Yakubovich–Popov lemma.

Human Interaction and RObotics [HIRO] Group **Apr 2025 – Present**

*Remote Undergraduate Researcher supervised by **Prof. Alessandro Roncone*** *University of Colorado Boulder*

- Developed procedurally generated alien terrain in Unity to support VR (Virtual Reality) based human-robot collaboration experiments.
- Designed and implemented puzzle-solving mechanisms enabling structured human–robot collaboration tasks in VR.

RINGZ Lab (Research in Networks and Groups) **Jul 2024 – Jan 2025**

*Undergraduate Researcher supervised by **Prof. Diego Gómez-Zará*** *University of Notre Dame*

- Developed a VR meeting room prototype using Unity and Photon Networking to create multi-user, immersive environments, improving virtual collaboration in distributed teams.
- Leveraged Generative AI and Large Language Models for better scene understanding and object manipulation of users in mixed reality applications.

Summer Undergraduate Research in Software Engineering **Aug 2023 – Jan 2024**

*Remote Undergraduate Researcher supervised by **Prof. Darko Marinov*** *University of Illinois Urbana-Champaign*

- Assisted in establishing a pipeline of clients libraries for testing compatibility and stability of client integrations to validate the partial translation.
- Improved modularity and compatibility across multiple Maven-based Java projects.

Skills

Programming: Python, C#, Matlab, C, C++, Java, SQL
Tools and Frameworks: Unity, ROS, Docker, Git, PyTorch
Engineering: RealSense, Altium, Keil
Languages: English (TOEFL: 106), French (intermediate), Chinese (native)

Other Experience

BoringMedia Intelligent Technology Co. — **Sept 2024 – Dec 2024**
Intern — *Shanghai, China*

- Implemented the validation algorithm by data labeling for emphasizing human factors and human-machine interaction design in cockpit systems.

School Of Computing Summer Workshop — **Jul 2023**
Visiting Student — *National University of Singapore*

- Developed a 2D game from scratch within a 3-week timeframe collaborating with teammates.
- Acquired proficiency in Unity and C# for game development.

Service & Awards

- **Machine Learning Course Coordinator**, University of Shanghai for Science and Technology (Fall 2023)
- **Sociolinguistics Course Coordinator**, University of Shanghai for Science and Technology (Spring 2023)
- **Academic Excellence Scholarship**, 2021-2022, University of Shanghai for Science and Technology